INTEGRAL RECONSTRUCTION AND QUANTITATIVE METRIC DEGENERATION OF K3 BUNDLE MODULI

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Abstract. We construct a compact moduli eightfold of rank-seven bundles on a K3 surface whose moving projective lines specialize to Schubert lines in a deepest Grassmannian fibre $\mathrm{Gr}(2,4)$. Both middle Schubert classes extend integrally. The fibre is primitive, has self-intersection six, and has an explicit degree-eight attachment modulo six. For the natural L^2 metric defined by harmonic deformation representatives, these same cycles have exact periods and volumes, ambient diameter asymptotics, and complete cotangent-Grassmannian rescaled limits. We also prove a global comparison theorem for compact K3 bundle-moduli contractions without bubbling, with complete convergent quiver charts whose complex moment-zero fibres contain simple representations. A principal-stratum path argument identifies the central harmonic completion, and finite quotient distances give a variational bound for the distortion of the actual contraction graph. The bound tends to zero through arbitrary adjacent chambers. Integral cochains on all geometric overlaps recover additive cohomology, while universal-index restrictions calculate middle-degree attachments from a marked complement ring. A mapping-torus example shows why local fibre rings and divisor classes do not determine the required monodromy.

1. Introduction

A contraction remembers an exceptional cycle in several ways. Its local quotient describes the cycle and its incidence with neighboring fibres. Its embedding in the compact space determines integral intersection and attachment data. A degenerating Kähler metric measures how that same cycle disappears in distance. The relation between these descriptions is especially concrete for moduli of bundles on a K3 surface: stable factors determine a quiver, while universal families identify the quiver's cycles inside integral cohomology.

This paper joins these descriptions for compact contractions without bubbling. The central example has a moving projective line and a deeper Grassmannian fibre $\mathrm{Gr}(2,4)$. The line specializes to a Schubert line. The two middle Schubert classes then reveal information absent from divisor restrictions: they prove integral primitivity and calculate an attachment modulo six. On these same cycles, the actual harmonic metric has exact periods and explicit ambient diameter asymptotics. A general comparison theorem controls the graph of the entire compact contraction.

1.1. The compact Grassmannian example.

The Mukai pairing is

$$(r,c,s)\cdot(r',c',s') = c\cdot c' - rs' - r's.$$

All cohomology and homology in the following statement have integral coefficients. The harmonic metric is the real, unnormalized Dolbeault L^2 metric on moduli.

Date: November 7, 2026.

2020 *Mathematics Subject Classification.* 14J60 (primary); 14J28, 53C26, 55N35 (secondary).

Key words and phrases. K3 surface, moduli of bundles, integral cohomology, degeneration, harmonic metric, Grassmannian, Gromov–Hausdorff convergence.

For a universal topological K-class $\mathcal{E}$ on $X \times M$, write

$$I_z = p_!(z^\vee \otimes \mathcal{E}), \qquad z \in K^0_{\mathrm{top}}(X),$$

where $p : X \times M \to M$ and $p_!$ is the topological K-theory pushforward. Thus z ranges over all integral topological classes, not only algebraic ones. The marked index-Chern classes on a complement are the restrictions of $c_k(I_z)$. The normalization used below tensors $\mathcal{E}$ by the inverse determinant of a specified rank-one index, as constructed in Section 7.

Theorem 1.1. *There is a projective K3 surface with*

$$\mathrm{NS}(X) = \mathbb{Z}H \oplus \mathbb{Z}D, \qquad H^2 = 2, \quad H \cdot D = 0, \quad D^2 = -44, \quad H \text{ ample},$$

and a unique H-stable spherical bundle A of vector $a = (3, D, -7)$. Put $b = (1, 0, 1)$, $w = a + b$, and $v = 2a + b = (7, 2D, -13)$. For sufficiently small generic $t \neq 0$, the entire moduli space $M_\pm$ of vector v, for polarization $H + tD$, is a smooth compact eightfold of $K3^{[4]}$ type consisting of slope-stable bundles. The two chamber spaces have projective symplectic contractions $\pi_\pm : M_\pm \to Z$ to a common normal target with the following properties.

(1) *The reduced singular locus is a normal sixfold T with one singular point o. Its smooth part parametrizes classes $A \oplus F$, where F is an H-stable bundle of vector w; o represents $2A \oplus \mathcal{O}_X$. The fibres of $\pi_\pm$ are points, projective lines, and $G = \mathrm{Gr}(2, 4)$, respectively. The auxiliary generic w-moduli resolves T, with exceptional fibre $\mathbb{P}^3$.*

(2) *A moving projective line specializes, in a proper reduced family, to a minimal Schubert line $\ell \subset G$. Every homology inclusion $H_k(G) \to H_k(M_\pm)$ is primitive, and $[G]^2 = 6$. In the positive chamber, let $h = c_1(\mathcal{U})$, $p = c_2(\mathcal{U})$ for the positive tautological quotient on G. Normalized universal indices satisfy*

$$c(I_z)|_G = (1 + h + p)^{\chi(A,z)}, \qquad \int_G h^4 = 2, \quad \int_G h^2 p = \int_G p^2 = 1.$$

For $V = M_+ \setminus G$, these formulas calculate $\kappa : H^8(V) \to \mathbb{Z}/6$ on all marked index-Chern generators, and

$$H^8(M_+) \simeq \{(u, N) \in H^8(V) \oplus \mathbb{Z} : \ N \equiv \kappa(u) \pmod 6\}.$$

The marked complement ring remains input to this calculation.

(3) *Put $\kappa_t = 88\pi|t|/7$. The minimal line and the moving fibre both have harmonic area $\pi\kappa_t$. Either Schubert plane has volume $(\pi\kappa_t)^2/2$, and G has volume $(\pi\kappa_t)^4/12$. Their ambient diameters satisfy*

$$\operatorname{diam} G = \frac{\pi}{\sqrt{2}}\sqrt{\kappa_t}(1 + O(|t|)), \qquad \operatorname{diam} \mathbb{P}^2 = \frac{\pi}{2}\sqrt{\kappa_t}(1 + O(|t|)).$$

The ambient spaces, rescaled by $\kappa_t^{-1/2}$ and based on G, converge in complete pointed Gromov–Hausdorff topology to the unit-level cotangent Grassmannian quotient. The harmonic metric on $Z \setminus T$ completes to Z; the graphs of $\pi_\pm$ satisfy Theorem 5.5's distortion bound and give compact Gromov–Hausdorff convergence to that completion.

There is no additional non-locally-free or positive-charge boundary in this compact problem.

The local incidence geometry of the Grassmannian is classical. The point of the theorem is its realization as the entire deep fibre of this bundle-only compact family, together with its integral attachment and actual harmonic geometry. The equal-dimensional incidence resolutions for the two signs are isomorphic over the affine cone; a nontrivial Grassmannian flop is not part of the statement.

1.2. **The general reconstruction and metric comparison.** The general hypotheses, stated in Section 2.2, require compact chamber spaces with a common central slope target, absence of every ideal boundary, and full convergent quiver charts with a module-simple point in each complex zero fibre. The local charts include repeated factors, tangent directions, inverse solution coverage and the actual classifying maps.

Two kinds of global information enter. For all-degree additive reconstruction, retain the geometric regular locus and the local classifying families, with their isomorphism loci. These determine the actual multiple overlaps and monodromy. Their integral cochain diagram computes the chamber cohomology by Proposition 3.2. For a compact core E, Theorem 3.1 instead uses its intersection map, calculated universal-index restrictions, and the marked ring of $M \setminus E$. Under the stated middle-degree vanishing hypotheses it computes the boundary homomorphism and an integral fibre product. These are two useful levels of reconstruction; neither recovers an unknown global complement from singularity types.

The metric result is quantitative in finite-model distances. For a sequence of prescribed accuracies we construct finite covers, positive short-distance scales ℓ_j, relative errors $\alpha_j \to 0$, and variational discrepancies $\beta_j(\theta) \to 0$ for each fixed j. If R_θ is the graph of the actual contraction, then

$$\operatorname{dis} R_\theta \leq \inf_{j:\theta\in\mathcal{P}_j} \left[\left(\alpha_j + \frac{\beta_j(\theta)}{\ell_j} \right) D_* + \beta_j(\theta) \right],$$

where D_* is constructed from a fixed central chart cover. Definitions (12)–(15) make all these quantities explicit. The estimate is uniform through stabilizer changes and arbitrary adjacent-chamber approaches. It is a variational modulus, rather than a universal power-law rate.

Three points in the proof deserve emphasis. First, no-charge compactness and local classifying compatibility put *whole* fibres in the analytic charts. Second, paths through lower strata can be approximated in length by paths in the principal stratum; density alone would not identify the harmonic completion. Third, first-exit estimates compare both ambient distance infima, including paths which might leave a local model. A short-distance correspondence lemma then assembles these local estimates without using a qualitative global convergence theorem.

1.3. **Relation to earlier work.** Existence and deformation type of the K3 moduli spaces come from Yoshioka [Yos00, Yos01]. The slope compactifications and the surface comparison with gauge theory are due to Greb–Toma and Greb–Sibley–Toma–Wentworth [GT17, GSTW21]. Quiver descriptions and wall geometry are developed in Arbarello–Saccà and Bayer–Macrì [AS18, BM14]. Grassmannian geometry in K3 moduli also appears in Markman's Brill–Noether duality [Mar01]; the present compact realization is proved from its own rank and charge classification.

The integral generation theorem is Markman's [Mar07], and cochain descent is classical. Our attachment theorem isolates the elementary relative argument that turns *calculated* index restrictions into a boundary homomorphism respecting every integral relation. The mod-six application uses both Schubert directions; it does not claim a new presentation of Grassmannian cohomology. Kaplan–Schedler [KS24] study compatibility and monodromy in gluing crepant resolutions. Here existing compact chamber spaces and their geometric classifying data are hypotheses; our cochain construction retains, rather than solves for, their monodromy.

The qualitative formality theorem of Bandiera–Manetti–Meazzini [BMM21] already applies to the common-slope direct sums used here because their automorphism groups are reductive. The controlled convergent symplectic coordinates are imported from [Wue26c]. Relative finite balancing, complete quotient continuity, normal slices and the harmonic comparison are imported from [Wue26a, Wue26b]; their hypotheses and normalizations are restated in Section 2. The complete cotangent model metric is classical [BG97]. Our metric contribution is the whole-fibre and principal-path argument, the explicit global graph bound, and its application to these identified compact cycles. Quantitative comparison results such as Lakzian–Sormani [LS13, LS20] place ambient distance discrepancies among their inputs; we derive the discrepancies here from finite quotient models. The established qualitative Ricci-flat degeneration theory [Tos09, Son14] provides context, but is not an input to Theorem 5.5.

Sections 3–5 prove the general statements. Sections 6–8 prove Theorem 1.1 on one compact family. Section 9 recovers the fourfold applications and uses the same two Schubert classes to exhibit the necessity of retaining monodromy.

2. Geometric data and harmonic coordinates

We begin by specifying the contraction problems under consideration. The hypotheses distinguish the geometry needed to identify a compact moduli space from the local estimates needed to control its metric. They will be verified, including the entire ideal boundary, for the eightfold constructed in Section 6.

2.1. **Conventions.**

For a projective K3 surface X, write

$$v(F) = \mathrm{ch}(F)\sqrt{\mathrm{td}_X} = (r, c, s), \qquad ((r, c, s), (r', c', s')) = c \cdot c' - rs' - r's.$$

Here and below ch denotes the Chern character. We fix the holomorphic determinant. For a smooth family of Ricci-flat Kähler forms ω_θ, set $\deg_\theta F = c_1(F) \cdot [\omega_\theta]$ and $\mu_\theta(F) = \deg_\theta F / \operatorname{rank} F$. The harmonic metric is the real part of the L^2 Hermitian pairing on harmonic $(0,1)$-deformations, without division by the volume of X. In particular it is half the real connection convention.

If $\mathcal{E}$ is a universal topological K-class and $p : X \times M \to M$, our Mukai homomorphism and family indices are

$$\theta_v(z) = \left[p_*(\mathrm{ch}(\mathcal{E})\sqrt{\mathrm{td}_X}\, z^\vee)\right]_2, \qquad I_z = p_!(z^\vee \otimes \mathcal{E}), \qquad \chi(F, z) = -(v(F), v(z)). \tag{1}$$

For θ_v the argument belongs to $v^\perp$; the notation $z^\vee$ reverses its degree-two component. A class on X in a family-index expression is pulled back without further notation. Base line bundles may change $\mathcal{E}$; they do not change $\theta_v|_{v^\perp}$. The Kähler form of the actual harmonic metric has class

$$[\Omega_\theta] = 2\pi^2 \theta_v(b_\theta), \qquad b_\theta = \left(0, [\omega_\theta], \frac{c_1(v) \cdot [\omega_\theta]}{\operatorname{rank} v}\right). \tag{2}$$

This is the fixed-determinant Chern–Weil formula [BS16, Proposition 4.3, (4.6)] in the real Dolbeault normalization. Indeed, with $\Delta(\mathcal{E}) = c_2(\mathcal{E}) - (r-1)c_1(\mathcal{E})^2/(2r)$, that formula reads $\omega_{\mathrm{WP}}/(2\pi^2) = 2p_*(\Delta(\mathcal{E})\omega_\theta)$. Our form is $\omega_{\mathrm{WP}}/2$, and expansion of (1) gives (2). The computation is valid for local twisted families because their endomorphism bundles are untwisted.

2.2. **The class of contractions.** Fix an ample central class $H = [\omega_0]$, a primitive positive-rank Mukai vector v, and the smooth finite-dimensional family ω_θ. We impose the following three hypotheses.

(G1) Compact chamber spaces. There are finitely many adjacent chambers and a connected normal projective variety Z. For all sufficiently small generic parameters, the full moduli space M_θ of slope-stable bundles is smooth and projective and is also the generic Gieseker moduli space. There is a proper surjective morphism $\pi_\theta : M_\theta \to Z$, constant as a complex morphism within each chamber. Points of Z are the H-polystable classes of vector v. Every chamber object is H-semistable, and π_θ assigns its H-graded class. The H-stable locus $U \subset Z$ is connected and dense, and π_θ is an isomorphism over U. Fix a finite Whitney stratification refining the stable-factor decomposition types. The projective and fixed-H gauge topologies on Z agree.

(G2) Absence of an ideal boundary. Every central ideal Hermitian–Yang–Mills object of vector v has zero lost charge, and every H-semistable sheaf of that vector is locally free. Thus surface gauge compactness, with the smoothly varying background forms, has no concentration.

(G3) Full convergent quotient models. For each central class

$$E_p = \bigoplus_i F_i \otimes \mathbb{C}^{m_i},$$

the full doubled Ext-quiver complex moment zero fibre contains a module-simple representation of the specified dimension vector. Here module-simple means that there is no nonzero proper submodule. All loops, multiplicities and trivial tangent directions are retained. The convergent symplectic coordinates give local classifying identifications compatible with the central graded class and the coarse analytic germ of Z. In these germs,

$$\pi_\theta f_{p,\theta} = f_{p,0} q_{p,\theta}, \tag{3}$$

where $q_{p,\theta}$ is the projective quotient map to the affine zero quotient. This is initially a statement about germs, not about the full preimage of a target neighborhood.

The analytic ingredients in (G3) have precise hypotheses. The common-slope normal form [Wue26c, Proposition 2.1] applies to locally free H-polystable bundles on a Ricci-flat K3, with arbitrary ranks and multiplicities. It is a convergent equivariant holomorphic change $\Psi(x) = x + O(\|x\|^3)$, $D\Psi = I + O(\|x\|^2)$, identifying the full obstruction ideal with the complex moment ideal. Together with the local harmonic reduction [Wue26b, Theorem 1.1 and Sections 3,5] and finite balancing [Wue26a, Theorem 2.1], it gives

$$(1 - e_p) g_{p,\zeta_p(\theta)} \;\le\; f_{p,\theta}^* g_\theta \;\le (1 + e_p) g_{p,\zeta_p(\theta)}, \qquad e_p = C_p(|\theta| + \|x\|^2). \tag{4}$$

The inequalities concern tangents to actual smooth quotient pieces. The local theorem also covers all sufficiently nearby actual solutions.

The finite quotient metric in (4) is induced by the central harmonic norm on $\mathrm{Ext}^1(E_p, E_p)$. Its real moment covector, paired by ordinary multiplicity trace, is

$$\zeta_{p,i}(\theta) = 2\pi\big(r_i \mu_\theta(v) - \deg_\theta F_i\big), \qquad \sum_i m_i \zeta_{p,i} = 0, \qquad r_i = \operatorname{rank} F_i. \tag{5}$$

The compact group is $\prod_i U(m_i)$, modulo the ineffective common scalar. The rank weights instead occur in the spectral mass $\mathrm{Vol}(\omega_0) \sum_i r_i \operatorname{tr}(\xi_i^* \xi_i)$, whose small quadratic form is $\|[\xi, x]\|^2$.

These are different pairings. Constants in (4) depend on the fixed background and its complementary elliptic gaps. They do not depend on ratios of positive finite eigenvalues or on a nearby stabilizer. Formal quadraticity alone would not give this estimate; the controlled convergent coordinates are essential.

2.3. **What the reconstruction retains.** Retain U as a geometric space with its interpretation by stable bundles and local classifying families. Retain also the quotient domains with their actual families in this same moduli problem. On two domains, their overlap is the locus of pairs representing isomorphic bundles. Injective coarse classifying maps identify its two projections with open subsets and give a unique transition map. We take cochains only after these overlap spaces and maps have been specified. In particular a bare cochain complex of U is not enough to define the overlaps.

These geometric maps retain stratum monodromy and all multiple compatibilities. Local fibre types alone do not supply them. Specialization of exceptional classes will be computed from a separate proper family in Section 6.5. For the smaller middle-degree calculation, the global datum is instead the marked ring $H^*(M \setminus E, \mathbb{Z})$, where E is the chosen compact exceptional core. This complement generally contains exceptional fibres over other strata. In our eightfold it is larger than $\pi^{-1}(U)$.

3. Integral attachment and cochain descent

The useful reduction in the middle degree is elementary once the correct integral restrictions have been calculated. We state it first, so that the later Grassmannian computation has a definite target: the homomorphism κ, not just its finite codomain.

3.1. **A calculated boundary homomorphism.** Let M be a closed oriented smooth $2n$-manifold and $E \subset M$ a compact tame subset. Suppose it has a closed oriented regular neighborhood W retracting onto E, with smooth boundary L. Put $V = M \setminus E$. Collars identify the overlap of V with an open neighborhood of E, up to homotopy, with L. Assume

$$\begin{aligned} H_n(E,\mathbb{Z}) = \Lambda \text{ is free}, \qquad & H^n(E,\mathbb{Z}) = \Lambda^*, \\ H^{n-1}(E,\mathbb{Z}) = H_{n-1}(E,\mathbb{Z}) = 0, \qquad & B : \Lambda \longrightarrow \Lambda^* \text{ is injective.} \end{aligned} \tag{6}$$

Here the dual identification is evaluation, and B is the middle intersection map in W. Let $C = \operatorname{coker} B$.

Theorem 3.1. *Suppose homogeneous integral classes $g_j \in H^*(M,\mathbb{Z})$ have degree-n monomials generating $H^n(M,\mathbb{Z})$. Supply the ring $A = H^*(V,\mathbb{Z})$ with the distinguished classes $a_j = g_j|_V$, and calculate $r_j = g_j|_E$. Then $H^{n-1}(L,\mathbb{Z}) = 0$, $H^n(L,\mathbb{Z}) = C$, and*

$$\kappa\big(P(a_j)\big) = [P(r_j)] \in C \tag{7}$$

is a well-defined homomorphism on all of A^n. It is the actual boundary restriction and gives

$$H^n(M,\mathbb{Z}) \;\simeq\; \{(u,z) \in A^n \oplus \Lambda^* :\; \kappa(u) = [z] \text{ in } C\}. \tag{8}$$

The isomorphism is restriction to V and W.

Proof. Poincaré–Lefschetz duality and the sequence of (W,L) give

$$0 \longrightarrow H^{n-1}(L,\mathbb{Z}) \longrightarrow \Lambda \xrightarrow{B} \Lambda^* \longrightarrow H^n(L,\mathbb{Z}) \longrightarrow 0.$$

This proves the boundary assertions. Excision also gives $H^{n+1}(M,V;\mathbb{Z}) = H_{n-1}(E,\mathbb{Z}) = 0$. Therefore restriction $H^n(M,\mathbb{Z}) \to A^n$ is onto, and the monomials in the a_j generate A^n.

It remains to check every relation, which is the point of the formula. If $P(a_j) = 0$, the class $P(g_j)$ comes from $H^n(M, V; \mathbb{Z}) = \Lambda$. Its restriction to W is the image of that same element under B. Hence $P(r_j) \in B\Lambda$, so (7) is independent of its polynomial representative. The two restrictions of each g_j agree on L; consequently κ is the boundary restriction. The Mayer–Vietoris sequence, with $H^{n-1}(L, \mathbb{Z}) = 0$, now identifies $H^n(M, \mathbb{Z})$ with the kernel of $(u, z) \mapsto \kappa(u) - [z]$. □

This is an integral fibre product. It retains the finite boundary group, saturation and extension information that would disappear after tensoring with $\mathbb{Q}$. It requires the marked complement ring; it does not calculate that ring from the singularity. A basis change in Λ gives the naturally isomorphic fibre product. Changing the generating set changes the formula, but not the homomorphism, because it is the same boundary restriction. Isotopic regular neighborhoods give the same result under collar identifications.

3.2. **Integral generators from universal indices.** For primitive-vector K3 moduli, the required global generators come from (1). Markman's theorem [Mar07, Theorem 1] states that Chern classes of the topological K-theory Künneth factors of a universal class generate integral cohomology. The integral Euler pairing on $K^0_{\mathrm{top}}(X)$ is perfect. Taking an Euler-dual integral basis therefore expresses those factors as the indices I_z. Thus the classes $c_k(I_z)$ generate $H^*(M, \mathbb{Z})$. The same theorem gives odd vanishing and torsion-freeness in this K3 setting.

Both qualifications matter: all topological K-classes are allowed, and the generation is integral. Algebraic indices alone, or rational Chern characters alone, would not justify Theorem 3.1. In a concrete exceptional family, a base-line ambiguity can often be removed by the determinant of a rank-one global index. Section 7 performs exactly this normalization before calculating κ.

3.3. **All degrees and all overlaps.** The general construction uses the geometric data of Section 2.3. Whole-fibre coverage, proved in Proposition 5.1, supplies a finite open cover $\mathcal{U} = (U_i)$ of M_θ by those domains, together with domains in the regular locus. Write $U_I = \bigcap_{i \in I} U_i$. Every intersection is expressed in any one of its coordinates by the actual classifying transitions.

Proposition 3.2. *Let*

$$K^{p,q}(\mathcal{U}) = \prod_{i_0 < \cdots < i_p} C^q(U_{i_0 \cdots i_p}; \mathbb{Z}).$$

With alternating restriction differential δ and total differential $D = \delta + (-1)^p d$, restriction induces a quasi-isomorphism

$$C^*(M_\theta; \mathbb{Z}) \longrightarrow \operatorname{Tot} K(\mathcal{U}). \tag{9}$$

Changing the cover, slices or family trivializations gives the same object in the derived category of integral complexes.

Proof. The coarse coordinate maps are unique on their overlaps and hence satisfy the cocycle identity on triples, and all its higher consequences. Thus $D^2 = 0$.

Use first the singular chains subordinate to $\mathcal{U}$. Barycentric subdivision and its usual chain homotopy show that their inclusion in all singular chains is a chain-homotopy equivalence. For each subordinate simplex, the set of indices of cover members containing its image is a nonempty simplex. Its augmented Čech complex is contractible. Taking the corresponding

products proves exactness of the augmented cochain resolution in the Čech direction. The Čech degree is bounded because the cover is finite; its filtration gives (9).

A refinement commutes with augmentation, so its total-complex map is a quasi-isomorphism. Two covers have a common intersection refinement. Different refinement assignments induce the same map in the derived category, since both commute with the invertible augmentation from $C^*(M_\theta;\mathbb{Z})$. Finally a change of slices or trivializations conjugates the overlap diagram by the unique coarse coordinate maps. Base-line twists, and different presentations of a universal gerbe, do not change those coarse maps. □

The descent assertion is classical; compare the open-cover construction in [Hat02, Section 4G]. Its application here uses the geometric identification of the domains and their overlaps, rather than replacing them by lists of fibre cohomology groups. The complex retains all multiple intersections and monodromy. It computes additive integral cohomology; no cup-product or homotopy-type assertion is being inferred from this additive statement. A finite cover gives a finite diagram of complexes, not automatically a finite matrix calculation. Compatible cellular models can make it finite when their maps are also supplied.

In particular, the formal descent argument does not solve the existence or compatibility problem for arbitrary local resolutions. The local-to-global theory of Kaplan–Schedler [KS24] treats such compatibility and monodromy questions in a broader setting. Here the compact projective chamber spaces are already part of (G1). Our task is to identify whole local fibres within them and to calculate the integral attachments on their universal-index classes.

4. Paths through the principal stratum

Before comparing compact spaces, we identify the central metric that will be their limit. Density of the stable locus is not sufficient: a dense Riemannian stratum can have a larger intrinsic distance. We prove equality of distances by perturbing paths and controlling the short joins between their pieces.

4.1. **Short paths in a fixed semialgebraic set.**

We use the following elementary consequence of semialgebraic triangulation and uniform monotonicity [Dri98, pp. 116,130,133].

Lemma 4.1. *A nonempty bounded connected semialgebraic set $S \subset \mathbb{R}^N$, not necessarily closed, has finite intrinsic diameter for Euclidean path length. More precisely, every two points can be joined in S by a continuous semialgebraic path of length at most a constant C_S.*

Proof. Triangulate the compact closure compatibly with S. Its inverse image is a finite union of relative open simplices. Choose their barycenters and connect two when one included simplex is a face of the other. The connecting segment lies in the larger open simplex except at its endpoint and stays in the union. The resulting finite incidence graph is connected. Otherwise one component and its complement would be relatively open and closed: a relative limit between included cells can only occur along a face incidence, which would connect the components.

Choose a spanning tree. Join each variable endpoint to its cell barycenter and then follow the tree between barycenters. Transport the resulting polygonal paths through the semialgebraic triangulation map. There are finitely many endpoint-cell choices, so this is a semialgebraic family of continuous paths. Cell decomposition and uniform finiteness, applied with the two endpoints as parameters, bound the number of monotonicity intervals of each coordinate by a common integer. Every coordinate stays in the bounded range of S. The sum of its coordinate variations therefore gives a uniform finite bound for path length. □

For a positive-dilation-invariant set, applying the lemma in its unit ball gives joining paths of length $C_S\rho$ in its radius-ρ ball. This is a statement about a fixed set; it does not assert a uniform constant for arbitrary degenerating families.

4.2. **The simple zero quotient.** Let W be a full doubled-quiver representation, with compact group K a connected product of unitary groups modulo an ineffective common scalar. Put

$$A_0 = \{\mu_{\mathbb{C}} = 0,\ \mu_{\mathbb{R}} = 0\}, \qquad Q_0 = A_0/K.$$

Assume that the complex zero fibre contains a module-simple representation. Write A_0^s for the balanced simple locus and $Q_0^s = A_0^s/K$. We use the finite intrinsic theorem [Wue26a, Section 5]: Q_0, with the infimum of finite-stratum path lengths, is a proper geodesic space whose metric topology is the quotient topology. We also use its full normal metric slice and normal-simple results [Wue26a, Sections 4,6]. These retain the complete normal representation and require the simple-zero hypothesis.

Lemma 4.2. *There is a constant C_W such that every two points of A_0^s of norm at most ρ can be joined in A_0^s, within that ball, by a path of length at most $C_W\rho$. At any point of Q_0, nearby simple points admit simple joining paths whose length tends to zero and which stay in any fixed sufficiently small neighborhood.*

Proof. Crawley–Boevey's theorem [CB01, Theorem 1.2], under the existence of a simple representation, makes the complex zero fibre irreducible and generically simple. Its simple quotient is a connected smooth open complex variety. Kempf–Ness identifies this with Q_0^s. The compact-group quotient $A_0^s \to Q_0^s$ is proper with connected fibres. A separation of its source would induce a separation of its target, by the closed-map property and connected fibres. Hence A_0^s is connected.

Simplicity is semialgebraic: for each proper subdimension vector, the existence of invariant subspaces is a polynomial incidence condition on a product of Grassmannians, followed by projection. There are finitely many such vectors. The balanced moment equations are real polynomial, and A_0^s is invariant under positive dilation. The map $x \mapsto x/\max(1, \|x\|)$ retracts it onto its unit-ball part. That part is connected and bounded, so Lemma 4.1 and scaling prove the first assertion. The zero-dimensional case uses constant paths.

For the second assertion take the full normal slice at a balanced representative of the chosen point. Its stabilizer is connected, and its full normal representation again has a module-simple point by the normal-simple theorem. Representatives of nearby points tend, after unitary adjustment, to zero in this slice. A closed zero-level representation has scalar stabilizer exactly when it is simple; the slice preserves this condition. Apply the first assertion in the full normal representation. The bounded metric comparison of the slice takes its $C\rho$ joins to joins of length at most $C'\rho$ in the ambient quotient. Their images lie in any prescribed fixed neighborhood for small ρ. □

Projection cannot increase the lengths of the balanced paths. Semialgebraic paths are piecewise smooth after finite subdivision, and within the simple stratum they may be smoothed with arbitrarily small additional length.

Proposition 4.3. *The intrinsic Riemannian distance of Q_0^s equals the restriction of the full finite-stratum distance of Q_0. Its metric completion is Q_0.*

Proof. One inequality follows because every simple path is admissible in the full quotient. For the reverse, take simple endpoints and a finite-stratum path of length within ϵ of their full distance. Truncate an arbitrarily small total length at its finitely many stratum junctions.

Cover the remaining compact smooth pieces by normal slice charts with relative squared-metric errors less than a chosen $\delta > 0$, and subdivide.

In a full slice the given stratum is $T \times \{0\}$, where the group acts trivially on T. Choose balanced simple normal points $(u_j, z_j) \to 0$. Translation in T removes u_j, without changing the moment map or stabilizer. Replace a piece $(u(t), 0)$ by $(u(t), z_j)$. Its product-metric speed is exactly the old speed. Comparing both curves with that same product metric increases length by at most $\sqrt{(1+\delta)/(1-\delta)}$. Their endpoints converge.

Use Lemma 4.2 to join the perturbed endpoints at each subdivision point. At an original singular junction, first choose a neighborhood where the joining cost is below an assigned tolerance, then truncate far enough toward the junction to place both endpoints there. Small simple perturbations remain there. The same construction joins to the original simple endpoints. There are finitely many joins; allocate their total cost below ϵ, and choose the chart errors so their total length inflation is below ϵ. The resulting simple path has length at most the full distance plus 3ϵ. Let $\epsilon \downarrow 0$.

The simple locus is dense in the quotient topology, hence in the finite metric topology. Properness gives completeness of Q_0, so it is the asserted completion. □

This argument explains why lower-dimensional strata create no new shortcuts between stable points: every such path can be approximated with its length, not merely with its endpoints.

5. Whole fibres and quantitative global distances

The local estimates have relative errors, whereas gluing distances introduces additive errors. We keep these separate. First we obtain whole-fibre charts and the central completion. We then compare short ambient pairs in both directions, and finally concatenate geodesics.

5.1. **Coverage and compatibility.**

Proposition 5.1. *Under* (G1)–(G3)*, local classifying charts can be chosen to contain the whole preimages of smaller target neighborhoods, for all sufficiently small generic parameters. For each prescribed relative accuracy they give finite covers with that accuracy. Their coarse transitions satisfy all cocycle identities, and if two squared-metric errors are $\epsilon_i, \epsilon_j < 1/2$, then*

$$\frac{1-\epsilon_i}{1+\epsilon_j} g_i \le h_{ij,\theta}^* g_j \le \frac{1+\epsilon_i}{1-\epsilon_j} g_i. \tag{10}$$

Proof. By (G2), a sequence of nearby chamber connections has a subsequence converging smoothly modulo gauge to a central polystable bundle. Indeed surface compactness first gives an ideal limit; the small-energy estimates and removability leave only positive integral lost charge, which (G2) excludes. The smoothly varying background metrics and determinant connections preserve these local estimates; see also [Wue26c, Lemma 3.2].

Suppose $\theta_n \to 0$ and $\pi_{\theta_n}(E_n) \to p$. Let E be a smooth gauge subsequential limit. Inverse coverage in the local solution chart at E gives balanced coefficients $x_n \to 0$, after compact-group adjustment. Equation (3) identifies their target classes with the affine invariants of x_n, which converge to the vertex at $[E]$. Since Z is Hausdorff, $[E] = p$.

Fix an outer chart at p. If no smaller target neighborhood and parameter ball had their whole preimages inside it, choose points outside the chart with targets tending to p and parameters tending to zero. The preceding conclusion puts them in its open actual solution neighborhood, a contradiction. Finite invariant properness bounds balanced representatives with bounded invariants and small levels. Shrinking the target neighborhood therefore puts

its whole preimage in any chosen inner coefficient ball. Conversely, the finite projective quotient is onto its affine target, and its small representatives classify actual nearby bundles. This gives coverage on both sides.

For a prescribed accuracy, shrink an outer chart separately at each p using (4). Extract a finite subcover of their inner target images, and only then shrink the common parameter ball. This order avoids any claim of a uniform complementary elliptic gap for separate stable backgrounds approaching a reducible limit. A chart at the reducible limit covers that region.

On overlaps put $h_{ij,\theta} = f_{j,\theta}^{-1} f_{i,\theta}$. These are the unique coarse identifications of the same holomorphic bundle classes, so they satisfy $h_{jk,\theta} h_{ij,\theta} = h_{ik,\theta}$. They are holomorphic in complex coordinates and smooth on smooth quotient pieces. Comparing each finite metric with the same actual harmonic metric gives (10). □

Universal families on overlapping domains can differ by a base line or a gerbe presentation. The coarse transition used above is unaffected. Proposition 5.1 also supplies the geometric cover used in Proposition 3.2.

5.2. **The central length space.** Give each central representation-type stratum its harmonic metric. Let d_0 be the infimum of the lengths of paths with finitely many smooth stratum pieces and finite one-sided endpoint lengths.

Proposition 5.2. *The function d_0 is a metric inducing the projective topology of Z. It makes Z a compact geodesic space and is the completion metric of the intrinsic harmonic metric on U. A finite upper bound $D_* \geq \operatorname{diam}(Z, d_0)$ can be constructed from one finite central chart cover and its coarse metric bounds.*

Proof. At zero level the model radius is the distance to the cone vertex. Small model neighborhoods thus connect nearby points at cost bounded by a fixed multiple of their radii. The local tensor comparison transfers these paths to Z. Connectedness and a finite chart cover make all distances finite.

There is also a lower bound for leaving a chart. With squared metric comparison between $1/2$ and 2, a path beginning at model radius $r < R$ and leaving the radius-R sublevel has length at least

$$(R - r)/\sqrt{2}. \tag{11}$$

The model radius is 1-Lipschitz, and the image of a closed sublevel is compact, so a first exit must meet its radial boundary. This applies to ambient paths, without assuming that their later portions remain in the chart. Local finite metric lower bounds, together with (11), separate distinct points. The short local paths give the reverse topological continuity. Thus d_0 induces precisely the topology of Z. Projective compactness gives metric compactness; the path definition makes it a length metric and hence a geodesic metric.

Apply the perturbation proof of Proposition 4.3 to a finite subdivision of a central actual path. Both the original and perturbed curves are compared with the same local model, using arbitrarily small relative errors. The simple joins have arbitrarily small total cost. Therefore the intrinsic distance on U is $d_0|_U$. Density and compactness identify its completion with Z.

For the last assertion choose a finite cover by connected inner radial neighborhoods with coarse comparison at most 2. Its intersection graph is connected. Choose an overlap point for each edge in a spanning tree. Within a chart, join its center and these points through its cone vertex, with length bounded by its radius times the coarse comparison factor. A simple tree path visits each chart at most once. The sum of these finitely many local bounds,

including the two endpoint joins, is a permissible D_*. No unknown global distance enters this construction. □

5.3. **Finite-model discrepancies.** Fix a central chart p, and write Q_ζ for its complete finite quotient, d_ζ for its intrinsic metric, r_ζ for the norm of a balanced representative, and $q_\zeta : Q_\zeta \to Q_0$ for the projective quotient map. For $r > 0$, define

$$b_{p,r}(\theta) = \sup_{\substack{q_\zeta(x)=z,\ q_\zeta(x')=z' \\ r_0(z),r_0(z')\le r}} |d_\zeta(x,x') - d_0(z,z')|, \tag{12}$$

$$a_{p,r}(\theta) = \sup_{\substack{q_\zeta(x)=z \\ r_0(z)\le r}} |r_\zeta(x) - r_0(z)|, \qquad \widetilde{a}_{p,r}(\theta) = \sup_{r_\zeta(x)\le r} |r_0(q_\zeta(x)) - r_\zeta(x)|, \tag{13}$$

where $\zeta = \zeta_p(\theta)$; the d_0 in (12) is the finite zero-model distance. We distinguish it from the central global distance by its arguments.

Lemma 5.3. *The three quantities in (12)–(13) are finite and tend to zero as $\theta \to 0$.*

Proof. Properness of the projective quotient and finite invariant properness bound balanced representatives uniformly for small levels and bounded target radii. The source-radius condition already gives bounded representatives in the other supremum. All relevant sets are compact.

If one convergence failed, take a contradicting sequence and convergent subsequences of its balanced representatives. Source and target have the same affine invariants. At zero level uniqueness of the balanced orbit identifies their limits. Continuity of radius proves the two radial assertions. Finite intrinsic parameter continuity [Wue26a, Theorem 5.1], applied to the two pairs of representatives, proves the distance assertion. There are finitely many parameter faces; passage to a face subsequence covers all approaches. □

These are variational functions of specified finite moment equations, norms and path distances. No semialgebraicity of those distances is required. The two radial functions serve different purposes: one bounds the preimage of a target ball, the other the image of a source ball.

5.4. **From short pairs to global distortion.** For a correspondence $R \subset X \times Y$, write

$$\operatorname{dis} R = \sup_{(x,y),(x',y')\in R} |d_X(x,x') - d_Y(y,y')|.$$

Lemma 5.4. *Let X, Y be nonempty compact length spaces and R a correspondence. Suppose $\ell > 0$, $\alpha, \beta \ge 0$, and*

$$d_X(x,x') \le \ell \implies d_Y(y,y') \le (1+\alpha) d_X(x,x') + \beta,$$
$$d_Y(y,y') \le \ell \implies d_X(x,x') \le (1+\alpha) d_Y(y,y') + \beta$$

for all related pairs. With $D = \operatorname{diam} X$ and $q = \alpha + \beta/\ell$, one has

$$\operatorname{dis} R \le qD + \beta, \qquad d_{\mathrm{GH}}(X,Y) \le \tfrac{1}{2}(qD+\beta), \qquad \operatorname{diam} Y \le (1+q)D + \beta. \tag{14}$$

Proof. Subdivide an X-geodesic of positive length $d_X(x,x')$ into $N = \lceil d_X(x,x')/\ell \rceil$ segments. Choose arbitrary related Y-points at the interior division points and the prescribed endpoints. Summing gives $d_Y(y,y') \le (1+q) d_X(x,x') + \beta$. For zero X-distance the local inequality gives this directly. Interchanging X, Y gives the reverse global inequality. If $d_Y \ge d_X$, their difference is at most $q d_X + \beta \le qD + \beta$. If $d_X \ge d_Y$, it is at most $q d_Y + \beta \le qD + \beta$.

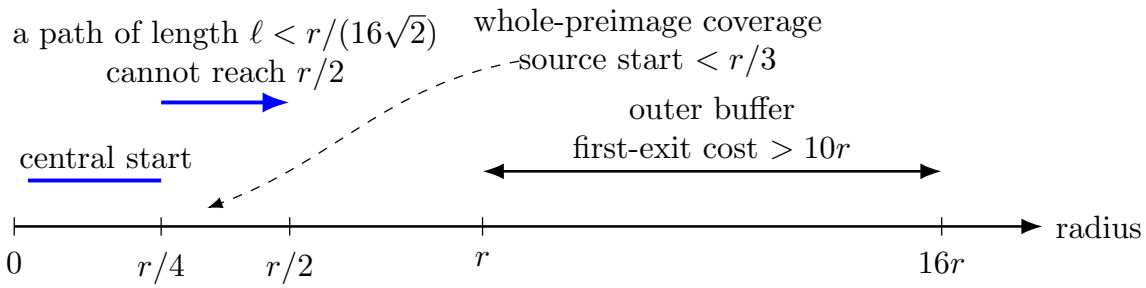


FIGURE 1. The radius buffers for one chart, schematically and not to scale. A short pair starts in the central $r/4$ ball and remains in the range where both finite discrepancies apply. The outer $16r$ chart excludes ambient shortcuts. The source start bound uses a_r; the reverse transfer of its endpoint uses $\widetilde{a}_r$.

This proves the distortion estimate, and the other assertions follow from the correspondence bound and the first global inequality. □

We now specify every quantity in the global bound. Choose $0 < \epsilon_j \downarrow 0$, with $\epsilon_j < 1/8$. At each central point choose an outer coefficient radius $16r$ where the central error in (4) is at most $\epsilon_j/2$. Extract a finite cover by central radius-$r/4$ images; write its radii as r_{ji}. After that choice, shrink a parameter neighborhood $\mathcal{P}_j$ so the nearby outer errors are at most ϵ_j, whole-preimage coverage holds, and

$$a_{i,r_{ji}},\ \widetilde{a}_{i,r_{ji}} < r_{ji}/16, \qquad b_{i,r_{ji}} < r_{ji}.$$

Choose any

$$0 < \ell_j < \min_i \frac{r_{ji}}{16\sqrt{2}}, \qquad \alpha_j = \sqrt{\frac{1+\epsilon_j}{1-\epsilon_j}} - 1, \tag{15}$$
$$\beta_j(\theta) = \sqrt{1+\epsilon_j}\, \max_i b_{i,r_{ji}}(\theta).$$

Theorem 5.5. *Under* (G1)–(G3), *let* R_θ *be the graph of* π_θ, *with the harmonic distance on* M_θ *and the completion distance on* Z. *Then*

$$\operatorname{dis} R_\theta \le \inf_{j:\,\theta\in\mathcal{P}_j} \left[\left(\alpha_j + \frac{\beta_j(\theta)}{\ell_j} \right) D_* + \beta_j(\theta) \right]. \tag{16}$$

For each fixed j, $\mathcal{P}_j$ *contains all sufficiently small generic parameters and* $\beta_j(\theta) \to 0$. *The right side tends to zero through arbitrary adjacent chambers. Consequently* (M_θ, d_θ) *converges in compact Gromov–Hausdorff topology to the harmonic completion* (Z, d_0).

Proof. Fix j and one of its charts, and abbreviate $r = r_{ji}$, $\epsilon = \epsilon_j$. First suppose both central endpoints have model radius at most r. Their finite central distance is at most $2r$, by joining through the cone vertex. Their finite source distance is at most $2r + b < 3r$, and source radii are at most $r + a < 17r/16$. Finite near-minimizers stay inside the radius-$16r$ chart, because radius is 1-Lipschitz. Their actual images give competitors of length less than $4r$. On the other hand, any actual path leaving that outer chart pays more than $10r$ before its first exit, by (11). The closed sublevel images and whole-preimage coverage justify this barrier

for ambient paths. Thus taking infima in both directions gives

$$\begin{aligned} d_\theta(x,x') &\le (1+\alpha_j)d_0(\pi_\theta x,\pi_\theta x') + \beta_j(\theta),\\ d_0(\pi_\theta x,\pi_\theta x') &\le (1+\alpha_j)d_\theta(x,x') + \beta_j(\theta). \end{aligned} \tag{17}$$

The relative factor compares the two squared-metric errors; the larger required additive factor is $\sqrt{1+\epsilon}\,b$.

It remains to put every short ambient pair in such a chart. Choose an inner radius-$r/4$ chart containing $\pi_\theta x$. Its entire source preimage has radius at most $5r/16 < r/3$. If the central distance to $\pi_\theta x'$ is at most ℓ_j, a central near-minimizer cannot reach radius $r/2$: its exit cost is at least $r/(4\sqrt{2}) > \ell_j$. Hence the other target endpoint is inside radius $r/2$. If instead the source distance is at most ℓ_j, the source exit cost from radius at most $r/3$ to radius $r/2$ is at least $r/(6\sqrt{2}) > \ell_j$. Thus x' has source radius less than $r/2$. The function $\widetilde{a}$ then puts its target inside radius r. In either case both inequalities (17) apply.

Use Lemma 5.4 with $X = (Z, d_0)$, $Y = (M_\theta, d_\theta)$, and the transposed graph of π_θ. Both spaces are compact length spaces and $\operatorname{diam} X \le D_*$ by Proposition 5.2. This proves the expression in brackets for each admissible j. Taking its infimum proves (16).

For fixed j, the finite discrepancy lemma gives $\beta_j(\theta) \to 0$. Therefore the limit superior of the bound is at most $\alpha_j D_*$. Now let $j \to \infty$. The correspondence estimate in (14) proves the stated convergence. □

The construction first fixes accuracy, then selects a finite cover, then takes the parameter limit. It gives an explicit variational modulus, with no assertion of a universal power law. In particular it does not use qualitative global Gromov–Hausdorff convergence as a premise.

6. A compact contraction with Grassmannian fibre

We now construct a family for which the general statements have two independent Schubert directions to detect. Its central repeated factor has multiplicity two, so its effective local group is $U(2)$. The construction includes the full compact moduli spaces, not only the Grassmannian quotient near one point.

6.1. The surface and the complete factor list.

Choose a K3 surface with

$$\mathrm{NS}(X) = \mathbb{Z}H \oplus \mathbb{Z}D, \qquad H^2 = 2, \quad H\cdot D = 0, \quad D^2 = -44. \tag{18}$$

Such a surface exists: the primitive embedding $H = e_1 + f_1$, $D = e_2 - 22f_2$ into two hyperbolic planes and Morrison's realization theorem [Mor84, Corollary 2.9(i)] give the full Néron–Severi lattice. A root would satisfy $h^2 - 22m^2 = -1$, impossible modulo eleven. We choose the positive component so that H is ample. Put

$$a = (3, D, -7), \quad b = (1, 0, 1), \quad w = a + b = (4, D, -6), \quad v = 2a + b = (7, 2D, -13). \tag{19}$$

Then

$$a^2 = b^2 = -2, \quad (a,b) = 4, \quad w^2 = 4, \quad v^2 = 6, \quad (a,w) = 2, \quad (a,v) = 0.$$

Lemma 6.1. *For central polystable sheaves or residual ideal bundles of total vector $a+(0,0,l)$, $w+(0,0,l)$, or $v+(0,0,l)$, with $l \ge 0$, the only possible stable-factor vector lists are*

$$\{a\}; \qquad \{w\}, \{a,b\}; \qquad \{v\}, \{a,w\}, \{a,a,b\},$$

respectively. In every case $l = 0$.

Proof. Every stable factor has vector (r, mD, s), since its H-slope is zero. Simplicity, Serre duality and Mukai Riemann–Roch give its square at least -2, hence

$$s \le \left\lfloor \frac{1 - 22m^2}{r} \right\rfloor.$$

For total vector $(R, MD, S + l)$, summation and Cauchy–Schwarz give

$$22 \sum_i \frac{m_i^2}{r_i} \le \sum_i \frac{1}{r_i} - S - l \le R - S, \qquad \left(\sum_i |m_i| \right)^2 \le \frac{R(R-S)}{22}. \tag{20}$$

For a, w the last bounds are $30/22, 40/22$. Since $\sum m_i = 1$, there is exactly one nonzero m_i, equal to one. For v the bound is $140/22 < 9$, and $\sum m_i = 2$; the nonzero entries are one two or two ones. In particular negative entries cannot occur.

For zero-m factors of total rank z, the maximal sum of third components is z, attained only by z rank-one factors with $s = 1$. For a, the maximal total third component with the nonzero factor of rank $r = 1, 2, 3$ is $\lfloor -21/r \rfloor + 3 - r = -19, -10, -7$. For w, the corresponding values at $r = 1, 2, 3, 4$ are $-18, -9, -6, -6$. These give precisely the stated lists.

For v, a single $m = 2$ factor of rank $r = 1, \dots, 7$ gives the bounds

$$-81, \ -39, \ -25, \ -19, \ -16, \ -14, \ -13.$$

For two $m = 1$ factors, order the ranks $r \le q$, $r + q \le 7$. The bound is $\lfloor -21/r \rfloor + \lfloor -21/q \rfloor + 7 - r - q$. All possibilities are

(r, q)	$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	$(1,5)$	$(1,6)$
bound	-37	-28	-25	-25	-25	-25
(r, q)	$(2,2)$	$(2,3)$	$(2,4)$	$(2,5)$	$(3,3)$	$(3,4)$
bound	-19	-16	-16	-16	-13	-13

Only $r = 7$ in the first case, and $(3, 3), (3, 4)$ in the second, survive. They give $v, 2a + b, a + w$. Every surviving bound already equals S. Thus $l = 0$ and each factor attains its bound. This exhausts all ranks and charges. □

The double dual of a slope-stable torsion-free sheaf is slope-stable. A positive discrepancy of length l lowers its Mukai square by $2rl$. For the vectors a, w, v, the resulting squares would be at most $-8, -4, -8$, respectively, contradicting the stable square bound. Thus stable sheaves of these vectors are bundles. The vector b gives $\mathcal{O}_X$. Extensions of the listed stable bundles are bundles, so every central semistable sheaf of these vectors is locally free.

6.2. **Existence and compactness.** Yoshioka's generic-polarization theorem [Yos00, Theorem 0.1] gives a stable object of vector a in a chamber adjoining H. A proper positive-rank class cannot have normalized first Chern class $D/3$. After avoiding the finite slope walls, generic Gieseker stability is therefore slope stability. Move the polarization to H inside this chamber. Each fixed subsheaf inequality becomes weak, so the object is H-semistable. Lemma 6.1 makes it stable, and the double-dual argument makes it locally free. Denote it by A. If two nonisomorphic H-stable bundles had vector a, their Hom and Ext^2 would vanish, contradicting Euler characteristic $-(a, a) = 2$. The same identity gives $\mathrm{Ext}^1(A, A) = 0$. Thus A is unique and spherical. Set $B = \mathcal{O}_X$.

The primitive vectors w, v similarly have compact generic moduli of dimensions six and eight, of deformation types $X^{[3]}$ and $X^{[4]}$ [Yos01, Appendix, Theorem 8.1]. For v, use large integral polarizations $mH \pm D$: the degree ∓ 88 is coprime to seven. For w, avoid the locally finite slope walls; equality of normalized first Chern classes would require $rD/4$ integral with

$0 < r < 4$, which is impossible. The double-dual argument applies at any polarization and makes every slope-stable object in these compact moduli locally free. Write $S_\pm$ for the sixfolds and $M_\pm$ for the eightfolds.

Finite wall theory [GT17, Theorem 6.6] permits a sufficiently small interval $H + tD$ with no wall except possibly $t = 0$. Every chamber object is H-semistable by the preceding limiting inequalities. Conversely every H-stable object persists in both adjacent chambers by openness and chamber constancy. The local models below show that $t = 0$ is an actual wall. Let ω_t be the smoothly varying Ricci-flat representatives of these classes.

The eightfold is fine: the algebraic K-class $\mathcal{O}_X + \mathcal{O}_p$ has index $\chi(E) + \operatorname{rank} E = -6 + 7 = 1$; the universal-family criterion applies [HL10, Theorem 4.6.5]. For w, algebraic Mukai divisibility is $\gcd(4, 44, 6) = 2$. Since $w^2 = 4 > 0$, the positive-square obstruction theorem of Mattei–Meinsma [MM25] gives Brauer order two. We use local or twisted families on $S_\pm$, without assuming an analytic trivialization of this obstruction. Topological universal classes remain available for the index calculations.

Finally Lemma 6.1 excludes positive-charge central ideal objects. Surface compactness and removability [GSTW21, Section 3.1], with the uniform small-energy estimates for the smooth family ω_t, give smooth subsequential gauge convergence on all of X as $t \to 0$. There are no positive-charge ideal objects at sufficiently nearby parameters either. Otherwise choose such parameters tending to zero. A bounded-energy subsequence has a fixed positive defect integer; compactness of the residual connections adds only nonnegative defect, contradicting the central list. This verifies the full no-boundary requirement, including nearby parameters, rather than just local freeness on the displayed fibres.

6.3. **The full nonabelian local model.** At $E_o = A \otimes \mathbb{C}^2 \oplus B$, both self-$\operatorname{Ext}^1$ spaces vanish and the cross dimension is four. Let $\mathbb{V} = \operatorname{Ext}^1(B, A)^* \simeq \mathbb{C}^4$, and use coordinates

$$Z_1 : \mathbb{V} \to \mathbb{C}^2, \qquad Y_1 : \mathbb{C}^2 \to \mathbb{V}.$$

The subscript distinguishes this linear map from the global target Z. The effective group is $U(2)$, and the equations are

$$Z_1 Y_1 = 0, \qquad Z_1 Z_1^* - Y_1^* Y_1 = \zeta I_2, \qquad \zeta = \frac{88\pi}{7} t. \tag{21}$$

Indeed (5) gives $2\pi(3\mu_t(v) - \deg_t A) = 88\pi t/7$; the B-vertex level is -2ζ. These are full convergent coordinates by the normal form and harmonic reduction stated in Section 2. Qualitative formality is already covered by Bandiera–Manetti–Meazzini [BMM21, Theorem 5.1]: the automorphism group is reductive, although the stable factors have different reduced Hilbert constants. The additional coordinate estimates, rather than formality alone, supply the relative harmonic comparison.

For $\zeta > 0$, Z_1 is surjective. Set $P = \ker Z_1$, $\mathcal{U}_P = \mathbb{V}/P$, and $N = Y_1 Z_1$. Then $\operatorname{im} N \subset P \subset \ker N$. Conversely such an incident pair determines Z_1, Y_1 up to $GL(2)$. Thus the quotient is

$$T^* \operatorname{Gr}(2, \mathbb{V}) = \{(N, P) : \operatorname{im} N \subset P \subset \ker N,\ \dim P = 2\} \longrightarrow \mathcal{N} = \{N : N^2 = 0,\ \operatorname{rank} N \le 2\}. \tag{22}$$

The cotangent zero section $G = \operatorname{Gr}(2, \mathbb{V})$ has $Y_1 = 0$, $Z_1 Z_1^* = \zeta I_2$, and coefficient norm squared 2ζ. For $\zeta < 0$, Y_1 is injective and $P = \operatorname{im} Y_1$ gives the same incidence resolution over $\mathcal{N}$. Thus this equal-dimensional Grassmannian model is not a nontrivial flop. Fibres over ranks two, one and zero are, respectively, a point, $\mathbb{P}^1$ and G.

For clarity we check normality of the affine target used here. The variety $\{Z_1 Y_1 = 0\} \subset \mathbb{C}^{16}$, over $\operatorname{rank} Z_1 = k$, has dimension $k(6 - k) + 2(4 - k) = 8 + 4k - k^2$, with maximum twelve at $k = 2$. Every lower-rank pair is a limit of rank-two pairs by perturbing Z_1, with Y_1 fixed,

among maps annihilating $\operatorname{im} Y_1$. It is an irreducible complete intersection of four equations. If either map has rank two, the differential of Z_1Y_1 is onto $\operatorname{End}(\mathbb{C}^2)$. The locus where both ranks are at most one has dimension at most nine. The complete intersection is therefore regular in codimension one and normal. Reductive invariants preserve normality, and the matrix invariant Y_1Z_1 identifies the quotient with $\mathcal{N}$.

The model at $A \oplus B$, for vector w, is the four-arrow $U(1)$ quotient with level $22\pi t$. It contracts $T^*\mathbb{P}^3$ to the rank-at-most-one square-zero cone. At $A \oplus F$, for an H-stable w-bundle F, the self-deformation space of F has dimension six and $\dim \operatorname{Ext}^1(F, A) = (a, w) = 2$. The local contraction is consequently a smooth sixfold factor times $T^*\mathbb{P}^1 \to \mathbb{C}^2/\{\pm 1\}$, with transverse level ζ.

The simple-zero condition is explicit. In (21) at zero level, take $Z_1 = (I_2\ 0)$, $Y_1 = (0\ I_2)^{\mathsf{T}}$. A subrepresentation with zero B-component has its A-component annihilated by all rows of Y_1, hence is zero. A nonzero B-component forces the full A-component by the columns of Z_1. The auxiliary model uses the analogous pair $(1, 0, 0, 0)$, $(0, 1, 0, 0)^{\mathsf{T}}$. The two-arrow transverse model has the same two distinct nonzero coordinates. Trivial tangent directions do not change simplicity. Small multiples in the convergent charts give actual H-stable bundles. Their loci are nonempty and dense in the irreducible chamber moduli.

6.4. **Whole fibres and normal targets.** The slope classifying theorem of Greb–Toma [GT17, Main Theorem, Theorem 5.5 and Remark 5.9] first gives maps to the common reduced slope images. For the auxiliary family the local classifying maps descend despite base-line changes. At this stage we do not assume normality of the image or its analytic identification with $\mathcal{N}$.

We verify whole-fibre coverage before making that identification. Suppose $t_n \to 0$ and a chamber bundle E_n has central graded object $2A \oplus B$. The no-charge result gives a smooth gauge limit E. In a Jordan–Hölder filtration choose its first inclusion and last quotient, pass to fixed factor types, and normalize these holomorphic maps separately to L^2-norm one in the converging gauges. Elliptic estimates give nonzero limiting maps. A nonzero map between stable equal-slope factors splits the corresponding factor in the polystable bundle E.

If the first and last factors differ, their limits split off A and B; the remaining vector a gives the unique A. If both are A, the inclusion i_n and quotient q_n satisfy $q_n i_n = 0$, still true after separate normalization and passage to the limit. With only one A-summand in E, both nonzero maps would factor through it and have nonzero composition. Thus there are at least two copies of A; the remaining vector is b. In all cases $E = 2A \oplus B$. The two-factor proof gives whole-fibre coverage for $A \oplus B$ and for $A \oplus F$ with F in a compact part of its stable locus.

We also need joint coverage when $F_n \to A \oplus B$. At fixed H, the surface comparison [GSTW21, Theorem 4.11 and Corollary 4.15] identifies the slope and Uhlenbeck topologies. Since charge is zero, F_n converges in gauge to $A \oplus B$. Orient the extensions of A and F_n in either chamber and normalize their two maps as above. Their limiting composition vanishes. If the limiting map involving $A \oplus B$ has a nonzero B-component, it splits off B and A. Otherwise zero composition forces two A-summands. The limit is again $2A \oplus B$. One fixed deep chart therefore covers these joint limits.

All actual fibres lie in these charts for sufficiently small parameters. Conversely every finite core has norm $O(\sqrt{|t|})$ and lies in the chart. They classify the same holomorphic bundles and central graded classes, so chamber constancy identifies their whole fibres.

Normalize the common reduced projective image. Connectedness of the fibres just identified and Stein factorization make this normalization bijective on points. Denote the resulting

targets by Z for v and T for w. Whole-fibre coverage and properness give saturated neighborhoods in the actual and finite resolutions. Pull back target holomorphic functions through the classifying charts and descend them using $\pi_*\mathcal{O} = \mathcal{O}$ for a proper birational map to a normal target. Doing this in both directions gives inverse germ maps: their composites agree on the dense stable locus and hence everywhere. Thus the contractions have exactly the analytic models above.

The target T has one singular point o, represented by $A \oplus B$; $S^\circ = T \setminus \{o\}$ is the stable w-moduli. The map $F \mapsto A \oplus F$ embeds it into Z. At o, fix a spectator A-copy in the $U(2)$ model. The equivariant Kuranishi and Darboux maps preserve this fixed locus, which is the $U(1)$ four-arrow representation. On invariants its inclusion imposes the two-by-two minors of N, giving the closed rank-at-most-one subcone of $\mathcal{N}$. These local embeddings glue because they classify the same direct sums. Properness makes $T \to Z$ a closed algebraic embedding. The singular support of $\mathcal{N}$ is its rank-at-most-one locus. Consequently T is precisely the *reduced* singular locus of Z.

We have proved projective symplectic contractions $\pi_\pm : M_\pm \to Z$ with fibres

central class	H-stable v	$A \oplus F,\ F \in S^\circ$	$2A \oplus B$
fibre	point	$\mathbb{P}^1$	$\mathrm{Gr}(2,4)$.

The auxiliary contractions $S_\pm \to T$ have fibre $\mathbb{P}^3$ over o. The symplectic forms agree with the quotient forms in the charts and with the common stable form elsewhere. Surface comparison gives the projective/gauge topology, while the earlier factor and charge argument excludes every further boundary. This verifies (G1)–(G3) for the eightfold.

6.5. **The actual extension and specialization.** Let $\mathcal{U}$ be the positive tautological quotient on $G = \mathrm{Gr}(2, \mathbb{V})$. In the positive chamber, the actual universal family restricts, up to a base line, to

$$0 \longrightarrow A \boxtimes \mathcal{U} \longrightarrow \mathcal{E}_G \longrightarrow B \boxtimes 1 \longrightarrow 0. \tag{23}$$

To see that this is the actual extension family, not just its linear approximation, observe that on $Y_1 = 0$ the central $\mathbb{C}^* \subset GL(2)$ gives Z_1 weight 1 and Y_1 weight -1. An equivariant holomorphic coordinate change tangent to the identity has no nonlinear term surviving on this axis in either output. It fixes the axis. The harmonic recursion is also linear there, since two one-direction extension terms have zero product. The quotient $\mathbb{V} \to \mathcal{U}$ therefore gives (23) itself.

Take $N_0 \neq 0$ with $N_0^2 = 0$, $\operatorname{rank} N_0 = 1$. Put $L_0 = \operatorname{im} N_0$, $K_0 = \ker N_0$. The entire fibre over τN_0, $\tau \neq 0$, is

$$\{P : L_0 \subset P \subset K_0,\ \dim P = 2\} \simeq \mathbb{P}^1.$$

Its closure is the product family

$$\mathbb{P}^1 \times \Delta \longrightarrow T^*G, \qquad (P, \tau) \longmapsto (\tau N_0, P).$$

It is proper and flat, with reduced central fibre a minimal Schubert line in G. The whole-fibre chart transfers it to the compact moduli. Thus the moving fibre class f specializes with multiplicity one to the line class ℓ:

$$f \longmapsto \ell. \tag{24}$$

The directions are parametrized by flags $L_0 \subset K_0 \subset \mathbb{V}$; including the plane gives $L_0 \subset P \subset K_0$. Over S°, projective extension transitions are in PGL_2, so their action on the fibre's integral cohomology is trivial. This does not trivialize the projective bundle.

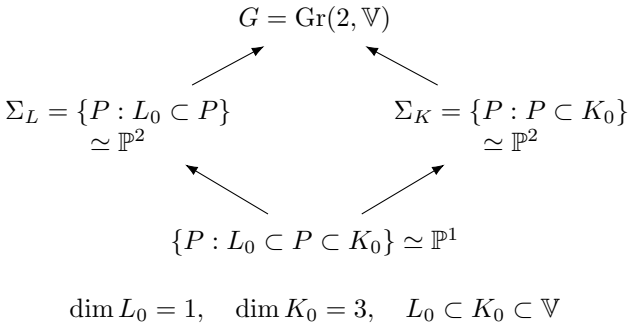


FIGURE 2. The Schubert line obtained from $N = \tau N_0$. For the fixed incident flag $L_0 \subset K_0$, the two Schubert planes meet along this line. The chosen flat family has this line as its reduced central fibre inside the larger fibre G. Arrows denote inclusions.

The normal bundle of G is T^*G. Writing a cotangent vector as a map $\mathbb{V}/P \to P$, the preimage of the rank-at-most-one locus has equation given by its 2×2 determinant. Its restriction line bundle is $\det P \otimes \det(\mathbb{V}/P)^* = \mathcal{O}_G(-2h)$, where $h = c_1(\mathcal{U})$. This records the incidence of the moving exceptional divisor with the deep core.

7. Schubert classes and the attachment modulo six

Work first in the positive chamber, with the actual extension (23). Write $h = c_1(\mathcal{U})$, $p = c_2(\mathcal{U})$. The integral Schubert presentation [Buc17, Section 3] is

$$H^*(G, \mathbb{Z}) = \mathbb{Z}[h, p]/(h^3 - 2hp,\ h^2p - p^2), \qquad |h| = 2, \quad |p| = 4, \tag{25}$$

with

$$\int_G h^4 = 2, \qquad \int_G h^2 p = \int_G p^2 = 1. \tag{26}$$

The second degree-four Schubert class is $q = h^2 - p$; thus $\int_G q^2 = 1$ and $\int_G pq = 0$. These are the classical two Schubert directions of $\mathrm{Gr}(2, 4)$. They are represented by planes of the two types: planes containing a fixed line, and planes contained in a fixed three-space.

7.1. **Removing the universal-family twist.** In the lattice embedding (18), put $D = e - 22f$. The integral Mukai vectors

$$x = (0, -f, 0), \qquad y = (0, 3f, 1)$$

are topological K-classes; algebraicity is not required. Their Euler pairings are

$$\chi(A, x) = 1, \quad \chi(B, x) = 0, \qquad \chi(A, y) = 0, \quad \chi(B, y) = 1.$$

If the restricted universal family differs from (23) by a line L, then $I_x|_G = \mathcal{U} \otimes L$ and $I_y|_G = L$. The global virtual rank of I_y is one. Tensor the universal class by $\det(I_y)^{-1}$; this removes L on G. For the resulting normalized indices,

$$I_z|_G = m_z[\mathcal{U}] + n_z[1], \qquad c(I_z|_G) = (1 + h + p)^{m_z}, \qquad m_z = \chi(A, z), \quad n_z = \chi(B, z). \tag{27}$$

The binomial expansion is integral for every integer m_z, including negative ones.

In particular $c_1(I_x)$ and $c_2(I_x)$ extend h, p integrally to M. Since h, p generate (25), restriction is onto in every degree. Duality makes every corresponding inclusion $H_k(G, \mathbb{Z}) \to H_k(M, \mathbb{Z})$ primitive. The class $[G]$ is primitive because the integral extension $c_2(I_x)^2$ evaluates

to one on it. This is stronger than the information from divisors: every divisor restricts to a multiple of h, whose fourth power has evaluation two.

7.2. **The boundary and complement.** The Plücker embedding identifies G with a smooth quadric fourfold in $\mathbb{P}^5$. The Euler and normal sequences give

$$c(TG) = \frac{(1+h)^6}{1+2h}, \qquad c_4(TG) = 3h^4 = 6p^2.$$

Since G is Lagrangian, its normal bundle is T^*G, so

$$[G]^2 = \int_G c_4(T^*G) = 6. \tag{28}$$

For the boundary $L = S(T^*G)$, the sphere-bundle Gysin sequence and (25) give

$$H^j(L,\mathbb{Z}) = \begin{cases} \mathbb{Z}, & j = 0,2,6,9,13,15, \\ \mathbb{Z}^2, & j = 4,11, \\ \mathbb{Z}/6, & j = 8, \\ 0, & \text{otherwise.} \end{cases} \tag{29}$$

The only nonzero Euler multiplication is from degree zero to degree eight, where it is multiplication by six. This is a smooth boundary of the resolved neighborhood; the target has a six-dimensional singular stratum through o, so it is not the link of an isolated singularity.

Put $V = M \setminus G$. Markman's theorem gives torsion-free even cohomology of M, and the primitive inclusions just proved make the relative sequence torsion-free. More explicitly, $H^k(M,V;\mathbb{Z}) = H_{16-k}(G,\mathbb{Z})$; its map to $H^k(M,\mathbb{Z})$ is the dual of restriction and is a primitive injection whenever nonzero. Odd groups of V vanish. The classical $X^{[4]}$ Betti numbers, obtained from Göttsche's formula [Göt90], therefore give

k	0	2	4	6	8	10	12	14	16
$\operatorname{rank} H^k(M)$	1	23	300	2852	19298	2852	300	23	1
$\operatorname{rank} H^k(V)$	1	23	300	2852	19297	2851	298	22	0.

The ranks removed in degrees $8, 10, 12, 14, 16$ are $1, 1, 2, 1, 1$. These ranks do not by themselves give the marked complement ring needed for attachment.

7.3. **The calculated congruence.** Apply Theorem 3.1 with $n = 8$, $\Lambda = \mathbb{Z}[G]$ and $B = (6)$. For a marked universal-index Chern monomial of degree eight on V, substitute (27), reduce by (25), and evaluate using (26), modulo six. This specifies a homomorphism $\kappa : H^8(V,\mathbb{Z}) \to \mathbb{Z}/6$ on all integral generators. In detail, for $m = m_z$,

$$\begin{aligned} c_1(I_z)|_G &= mh, \\ c_2(I_z)|_G &= mp + \binom{m}{2} h^2, \\ c_3(I_z)|_G &= \frac{m(m^2-1)}{3}\, hp, \\ c_4(I_z)|_G &= \frac{m^2(m^2-1)}{12}\, p^2. \end{aligned} \tag{30}$$

The displayed coefficients are integer-valued for every integer m. The ring relations give the rule for all products, including products involving different indices. The all-relations argument in Theorem 3.1 proves that this calculation depends only on the specified complement class.

We obtain the explicit integral reconstruction

$$H^8(M,\mathbb{Z}) \simeq \{(u,N) \in H^8(V,\mathbb{Z}) \oplus \mathbb{Z} : N \equiv \kappa(u) \pmod 6\}. \tag{31}$$

Moreover $\kappa(c_2(I_x)^2|_V) = 1$, so κ is onto. The higher index class supplies the odd evaluation which fourth powers of divisor classes cannot supply. Both Schubert directions, and their integral attachment, are thus visible in the same compact moduli family.

For the negative chamber the pure extension axis instead restricts the normalized universal class to $A \boxtimes \mathcal{S} + B$, where $\mathcal{S}$ is the tautological subbundle in the image-of-Y_1 description. Its Chern classes $c_1(\mathcal{S}) = -h$, $c_2(\mathcal{S}) = q = h^2 - p$ again generate the full integral ring. The same duality argument proves all primitive inclusions in that chamber as well.

The core is also genuinely different from a toric core component. Indeed $\mathrm{Pic}(G) = \mathbb{Z}h$, so a connected automorphism preserves the Plücker embedding and acts through the projective orthogonal group of its smooth quadric. Its maximal torus has dimension three. A dense torus action on a toric fourfold would require dimension four. This observation is not needed for (31); the two independent Schubert classes already explain the extra integral information.

8. Harmonic geometry of the same cycles

The integral identification now determines exact metric periods. Ambient diameters require an additional argument, because a shortest path between points of an exceptional cycle may leave it.

8.1. **Periods and primitive curve classes.** For $z \in v^\perp$, the normalized extension gives

$$\theta_v(z)|_G = -(a,z)h.$$

Thus a positive minimal line ℓ represents the functional $-a$ on $v^\perp$. Since $(a,v) = 0$, its Beauville–Bogomolov square is -2. The integral vector $z_0 = (0,-7f,-2) = x - 2y$ satisfies $(v,z_0) = 0$, $(a,z_0) = -1$; its Mukai class restricts to h, proving primitivity of ℓ directly. The proper family (24), rather than divisor pairings alone, identifies the moving fibre class with this line class.

For $t > 0$, put

$$\kappa_t = \frac{88\pi t}{7}, \qquad b_t = (0, H + tD, -88t/7). \tag{32}$$

We have $(a,b_t) = -44t/7$, so (2) gives the exact formulas

$$\begin{aligned} \mathrm{Area}_{g_t}(\ell) &= \mathrm{Area}_{g_t}(f) = \pi\kappa_t, \\ \mathrm{Vol}_{g_t}(\mathbb{P}^2) &= \frac{(\pi\kappa_t)^2}{2}, \qquad \mathrm{Vol}_{g_t}(G) = \frac{(\pi\kappa_t)^4}{12}. \end{aligned} \tag{33}$$

Either Schubert plane type is allowed. Its h^2-integral is one; the h^4-integral on G is two. These are Kähler volume identities on the actual holomorphic cycles, not a claim that the induced metric is exactly homogeneous. For $t < 0$, set $\kappa_t = 88\pi|t|/7$ and reverse the extension convention. The effective line functional is then $+a$ in that chamber's Mukai marking, and the positive formulas are unchanged.

8.2. **A distance-preserving zero section.** Consider the positive-level complete quotient

$$Z_1 Y_1 = 0, \qquad Z_1 Z_1^* - Y_1^* Y_1 = \kappa I_2$$

with the flat Hilbert–Schmidt metric. Let g_{Gr} denote the homogeneous Grassmannian metric with minimal line area π.

Lemma 8.1. *The projection to* $\ker Z_1$ *is a 1-Lipschitz retraction of the quotient onto its zero section with metric* κg_{Gr}*. Consequently the ambient distances between core points equal their intrinsic Grassmannian distances. In the unit normalization,*

$$\operatorname{diam} G = \frac{\pi}{\sqrt{2}}, \qquad \operatorname{diam} \mathbb{P}^2 = \frac{\pi}{2}$$

for either Schubert plane, with ambient distances in the complete quotient.

Proof. The real moment equation gives $Z_1 Z_1^* \geq \kappa I_2$. Let P_{row} be the orthogonal projector onto the row space of Z_1. The horizontal derivative of its row-space orthonormal frame has the form $(I - P_{\mathrm{row}})\delta Z_1^* (Z_1 Z_1^*)^{-1/2}$. The equivalent kernel description therefore satisfies

$$\kappa \, |d(\ker Z_1)|^2_{g_{\mathrm{Gr}}} \leq \|\delta Z_1\|^2 \leq \|\delta Z_1\|^2 + \|\delta Y_1\|^2.$$

The projection kills vertical orbit directions, so this bound survives minimization defining the quotient norm. On $Y_1 = 0$, the equation $Z_1 Z_1^* = \kappa I_2$ gives the Stiefel quotient metric κg_{Gr}. The retraction is the identity there; the inclusion and retraction prove equality of the two distances.

For two planes, use principal vectors to rotate the two orthogonal blocks through angles $\theta_1, \theta_2 \in [0, \pi/2]$. The horizontal frame path has length $\sqrt{\theta_1^2 + \theta_2^2} \leq \pi/\sqrt{2}$. For orthogonal planes, each column of any horizontal frame lift travels spherical distance at least $\pi/2$. Minkowski's integral inequality bounds its Hilbert–Schmidt length below by $\pi/\sqrt{2}$, proving sharpness. For either Schubert plane all planes share a fixed line or lie in a fixed three-space; one principal angle vanishes. The same argument gives diameter $\pi/2$ and the induced line-area-π Fubini–Study metric. □

The complete cotangent Grassmannian metric is classical [BG97, Theorem 1 and Section 2.2]. The retraction argument above verifies the ambient-distance statement needed for its transfer to the compact moduli.

Proposition 8.2. *In the actual eightfold, for either chamber,*

$$\operatorname{diam}_{M,g_t} G = \frac{\pi}{\sqrt{2}} \sqrt{\kappa_t}\,(1 + O(|t|)), \qquad \operatorname{diam}_{M,g_t} \mathbb{P}^2 = \frac{\pi}{2} \sqrt{\kappa_t}\,(1 + O(|t|)). \tag{34}$$

After distances are divided by $\sqrt{\kappa_t}$*, the ambient spaces based on* G *converge in complete pointed Gromov–Hausdorff topology to the unit-level* $T^* \operatorname{Gr}(2,4)$ *quotient. Along compact subsets of* S°*, the moving fibre has transverse Eguchi–Hanson limit and*

$$\operatorname{diam}_{M,g_t} f / \sqrt{\kappa_t} \longrightarrow \pi/2$$

uniformly.

Proof. The deepest finite core has coefficient norm squared $2\kappa_t$. Finite competitors have length $O(\sqrt{\kappa_t})$, and their actual images have that length bound. The first-exit cost from a fixed larger chart stays positive. An actual near-minimizer cannot leave it; the 1-Lipschitz finite radius and coarse metric bound then keep such a path in coefficient radius $O(\sqrt{\kappa_t})$. On this smaller region (4) has error $O(|t|)$. It therefore compares both ambient distance infima, not merely the tensors on the core. Lemma 8.1 gives (34).

After rescaling, fixed chart domains exhaust the entire finite quotient. Exit barriers diverge, relative errors tend to zero on every bounded model ball, and inverse coverage supplies the whole corresponding actual ball. Finite intrinsic continuity then gives the complete pointed limit. The unitary symmetry is transitive on the core, so any choice of core basepoint gives the same pointed limit up to isometry.

On a compact subset of S°, use finitely many full $\mathbb{C}^6$ times two-arrow quotient charts. Their normal zero sections are the moving projective lines, with the same level κ_t. The identical projection and first-exit argument gives the stated uniform diameter and transverse Eguchi–Hanson limit. The full ambient pointed limit also retains the Euclidean $\mathbb{C}^6$ factor. □

8.3. **Successive scales with moving centers.** We state the precise rule when a basepoint approaches a deeper stratum; a compact-stratum estimate alone does not cover that case. The rule applies to every contraction satisfying (G1)–(G3).

Fix a central chart at p, numbers $\delta_n \downarrow 0$, and generic parameters $\theta_n \to 0$. Suppose

$$\nu_n = \zeta_p(\theta_n)/\delta_n^2 \longrightarrow \nu, \qquad x_n \longrightarrow x$$

for balanced representatives x_n at level ν_n, after unitary adjustment. Then

$$(M_{\theta_n}, \delta_n^{-1} d_{\theta_n}, f_{p,\theta_n}([\delta_n x_n])) \longrightarrow (Q_\nu, d_\nu, [x]) \tag{35}$$

in complete pointed Gromov–Hausdorff topology.

For a further scale, retain a limiting representative x_* and its compact stabilizer K_*. Write its full normal representation as $T \oplus W'$, with trivial action on T. In the common slice suppose the current representatives are $(u_n, z_n) \to 0$. For $\sigma_n \downarrow 0$, require

$$\frac{(\nu_n - \nu)|_{\operatorname{Lie} K_*}}{\sigma_n^2} \longrightarrow \lambda, \qquad \frac{z_n}{\sigma_n} \longrightarrow z_*. \tag{36}$$

At physical scale $\delta_n \sigma_n$, the complete pointed limit is

$$T \times (W' /\!/\!/_\lambda K_*), \qquad \text{based at } (0, [z_*]).$$

The notation $/\!/\!/_\lambda$ means complex moment zero and real moment λ. Tangential motion u_n need only tend to zero; it need not be comparable with σ_n. This rule can be repeated for every fixed finite number of stages, retaining each stabilizer, full normal representation, normalized level and normal basepoint.

For completeness, the finite parameter and normal-slice results are [Wue26a, Sections 4–6]; their actual harmonic transfer is [Wue26c, Theorem 8.1]. The proof of that transfer uses the same three facts used above: relative errors on every bounded model region, divergent rescaled exit costs, and inverse coverage of entire balls. Full normal slices preserve the simple-zero condition, and tangential translation removes u_n. Errors compose multiplicatively and are not divided by the next smaller scale. Distance scaling is the inverse physical scale; tensor scaling is its inverse square. These arguments prove (35) and the finite iteration under precisely (36).

In the eightfold, a rank-one point of the zero cone has full normal model $\mathbb{C}^6$ times the A_1 quotient. Thus approaching the deepest point is handled by the $U(2)$ model first, followed, when appropriate, by this full slice. No single limit is assigned to oscillating normalized data, and no infinite iteration is asserted.

Proof of Theorem 1.1. Section 6 constructs the surface, the unique spherical bundle, both compact chamber spaces and their common normal target. Lemma 6.1 and the compactness argument exclude all additional ideal boundary; the whole-fibre and target arguments identify every stratum and fibre. The proper family in Section 6.5 proves reduced integral specialization. Section 7 gives the primitive inclusions, self-intersection and calculated mod-six attachment. Equations (33) and (34), with Proposition 8.2, give the cycle metric claims. The construction verifies (G1)–(G3), so Proposition 5.2 and Theorem 5.5 identify the harmonic completion and control the full compact contraction graph. □

9. Fourfold examples and the role of monodromy

9.1. **The isolated and interacting plane contractions.** The compact fourfolds in [Wue26c] provide two further applications of the general theorem. We verify its local simple-zero hypothesis explicitly and record the additional global metric conclusion; their construction and detailed cycle geometry remain in that paper.

For the isolated example, the lattice is $\mathrm{NS}(X) = \mathbb{Z}H \oplus \mathbb{Z}D$ with Gram matrix $\mathrm{diag}(2, -22)$, and

$$a = (1, 0, 1), \quad b = (2, D, -5), \quad v = a + b.$$

The compact chamber spaces have a single $\mathbb{P}^2$ fibre. The complete factor list and no-charge theorem [Wue26c, Section 3] verify (G1)–(G2). The full deep representation is $\mathbb{C}^3 \oplus (\mathbb{C}^3)^*$ under $U(1)$, with complex equation $y(x) = 0$. The point $x = e_1, y = e_2^*$ is module-simple, as both arrow directions between the two one-dimensional vertices are nonzero. At stable points the effective group is trivial. The convergent charts and classifying diagram verify (G3).

For the interacting example, the lattice matrix and vectors are

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -6 & -4 \\ 0 & -4 & -8 \end{pmatrix}, \qquad a_0 = (1, 0, 1), \quad a_1 = (1, D_1, -2), \quad a_2 = (1, D_2, -3), \quad v = a_0 + a_1 + a_2.$$

For $[\omega] = H + sD_1 + tD_2$, the levels are

$$\zeta(s, t) = \frac{2\pi}{3} \begin{pmatrix} -10 & -12 \\ 8 & 0 \\ 2 & 12 \end{pmatrix} \begin{pmatrix} s \\ t \end{pmatrix}.$$

All six chamber spaces, their common normal target, complete factor and charge lists, and whole fibres are constructed in [Wue26c, Sections 4–5]. They verify (G1)–(G2). At the deepest point the full effective group is $U(1)^3/U(1)$, with paired arrow dimensions $1, 2, 1$. In the coordinates of that paper,

$$zz' - \varphi(u) = ww' + \varphi(u) = 0$$

has the solution $z = z' = w = 1,\ w' = -1,\ u = \varphi = (1, 0)$. Its directed support is strongly connected, which for one-dimensional vertices forbids a proper submodule. Along the moving stratum the full representation is $\mathbb{C}^2$ times the two-arrow $U(1)$ model, which has the simple point e_1, e_2^*; stable centers are smooth. Together with the convergent classifying diagrams this verifies (G3) on every stratum.

Theorem 5.5 consequently gives graph distortion tending to zero in both examples, with the variational bound (16). For the interacting family this holds through all chamber directions and arbitrary ratios of positive scales. It supplements the qualitative global convergence of [Wue26c] by a proof from the finite quotient distances.

Their integral formulas are instances of Theorem 3.1, with the explicit normalized index restrictions already proved in [Wue26c, Proposition 6.1]. For the plane, $B = (3)$ and

$$\kappa(c_2(I_z)|_V) = \binom{m_z}{2}, \qquad \kappa(c_1(I_z)c_1(I_{z'})|_V) = m_z m_{z'} \pmod 3.$$

Here m_z is the Euler pairing with the extension subobject. In the two-plane chamber, $E = P_0 \cup P_2$ and

$$B = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \qquad \operatorname{coker} B \simeq \mathbb{Z}/8, \qquad (n_0, n_2) \longmapsto n_0 - 3n_2.$$

With $m_{i,z} = \chi(a_i, z)$, the boundary map is

$$\kappa(c_2(I_z)|_V) = \binom{m_{0,z}}{2} - 3\binom{m_{2,z}}{2},$$
$$\kappa(c_1(I_z)c_1(I_{z'})|_V) = m_{0,z}m_{0,z'} - 3m_{2,z}m_{2,z'} \pmod 8.$$

These formulas use the marked complement ring and respect all its relations. The mod-eight statement concerns this two-plane chamber. The other four configurations are not assigned its global formula.

The proper moving-line family specializes as $f = c_0 + c_2$, with both multiplicities one. For levels $\zeta_0 = a > 0, \zeta_2 = b > 0$, the exact areas are $\pi a, \pi b, \pi(a+b)$; the plane volumes are $(\pi a)^2/2, (\pi b)^2/2$. The ambient collapse and complete separated-scale limits are those of [Wue26c, Section 8], with unchanged metric normalization. Thus the general global comparison and the explicit attachment refer to those same cycles.

9.2. **The Schubert involution.** The following topological example explains why the geometric transition information in Section 2.3 cannot be replaced by local fibre rings and divisor classes.

Choose a nondegenerate symmetric complex bilinear form on $\mathbb{C}^4$. Orthogonal complementation defines a holomorphic involution $\delta(P) = P^\perp$ of $G = \mathrm{Gr}(2,4)$. For its tautological subbundle $\mathcal{S}$ and quotient $\mathcal{U}$, one has $\delta^*\mathcal{S} \simeq \mathcal{U}^*$. Consequently

$$\delta^* h = h, \qquad \delta^* p = q, \qquad \delta^* q = p.$$

Form the compact smooth mapping tori T_1 and T_δ over S^1. Both have the same local product geometry, integral fibre ring, and distinguished divisor class h. Their degree-four cohomology is different:

$$H^4(T_1, \mathbb{Z}) = \mathbb{Z}^2, \qquad H^4(T_\delta, \mathbb{Z}) = \mathbb{Z}. \tag{37}$$

Indeed $H^3(G, \mathbb{Z}) = 0$, so the Wang sequence identifies degree-four cohomology with the invariant subgroup of $H^4(G, \mathbb{Z}) = \mathbb{Z}p \oplus \mathbb{Z}q$. For the identity it is all of that group; for δ it is $\mathbb{Z}(p+q)$.

This is a topological necessity example, not a claim that the involution occurs in the K3 family above. It shows that forgetting monodromy can change even a Betti number. The use of all multiple intersections has a further purpose: cochains retain their restriction maps and higher compatibility, which fibre cohomology alone generally discards. Theorem 3.1 gives a smaller calculation only under its explicit vanishing, intersection and generator hypotheses.

The results concern compact K3 bundle-moduli contractions without bubbling. Extending them to positive-charge strata requires new local geometric and metric inputs. The reconstruction of underlying degenerating manifolds and unrestricted Hermitian–Yang–Mills dynamics are likewise separate questions.

References

[AS18] E. Arbarello and G. Saccà, *Singularities of moduli spaces of sheaves on K3 surfaces and Nakajima quiver varieties*, Adv. Math. **329** (2018), 649–703. doi:10.1016/j.aim.2018.02.003.

[BMM21] R. Bandiera, M. Manetti and F. Meazzini, *Formality conjecture for minimal surfaces of Kodaira dimension 0*, Compos. Math. **157** (2021), 215–235. doi:10.1112/S0010437X20007605.

[BM14] A. Bayer and E. Macrì, *MMP for moduli of sheaves on K3s via wall-crossing: nef and movable cones, Lagrangian fibrations*, Invent. Math. **198** (2014), 505–590. doi:10.1007/s00222-014-0501-8.

[BG97] O. Biquard and P. Gauduchon, *Hyperkähler metrics on cotangent bundles of Hermitian symmetric spaces*, in *Geometry and Physics (Aarhus, 1995)*, J. E. Andersen, J. Dupont, H. Pedersen and A. Swann, eds., Lecture Notes in Pure and Appl. Math. **184**, Marcel Dekker, New York, 1997, 287–298. Author text.

[BS16] I. Biswas and G. Schumacher, *The Weil-Petersson current for moduli of vector bundles and applications to orbifolds*, Ann. Fac. Sci. Toulouse Math. (6) **25** (2016), no. 4, 895–917. doi:10.5802/afst.1514.

[Buc17] A. S. Buch, *Notes on Grassmannians*, class notes, Rutgers University, 4 September 2017. Author text.

[CB01] W. Crawley-Boevey, *Geometry of the moment map for representations of quivers*, Compos. Math. **126** (2001), 257–293. doi:10.1023/A:1017558904030.

[Dri98] L. van den Dries, *O-minimal structures and real analytic geometry*, Current Developments in Mathematics **1998**, no. 1, International Press, 105–152. doi:10.4310/CDM.1998.v1998.n1.a4.

[Göt90] L. Göttsche, *The Betti numbers of the Hilbert scheme of points on a smooth projective surface*, Math. Ann. **286** (1990), 193–207. doi:10.1007/BF01453572.

[GSTW21] D. Greb, B. Sibley, M. Toma and R. Wentworth, *Complex algebraic compactifications of the moduli space of Hermitian–Yang–Mills connections on a projective manifold*, Geom. Topol. **25** (2021), 1719–1818. doi:10.2140/gt.2021.25.1719.

[GT17] D. Greb and M. Toma, *Compact moduli spaces for slope-semistable sheaves*, Algebr. Geom. **4** (2017), no. 1, 40–78. doi:10.14231/AG-2017-003.

[Hat02] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002. Author text.

[HL10] D. Huybrechts and M. Lehn, *The Geometry of Moduli Spaces of Sheaves*, 2nd ed., Cambridge Mathematical Library, Cambridge University Press, Cambridge, 2010. doi:10.1017/CBO9780511711985.

[KS24] D. Kaplan and T. Schedler, *Crepant resolutions of stratified varieties via gluing*, Internat. Math. Res. Notices **2024**, no. 17, 12161–12200. doi:10.1093/imrn/rnae135.

[LS13] S. Lakzian and C. Sormani, *Smooth convergence away from singular sets*, Comm. Anal. Geom. **21** (2013), no. 1, 39–104. doi:10.4310/CAG.2013.v21.n1.a2.

[LS20] S. Lakzian and C. Sormani, *Errata to "Smooth convergence away from singular sets"*, Comm. Anal. Geom. **28** (2020), no. 7, 1755–1772. doi:10.4310/CAG.2020.v28.n7.e1.

[Mar01] E. Markman, *Brill–Noether duality for moduli spaces of sheaves on K3 surfaces*, J. Algebraic Geom. **10** (2001), no. 4, 623–694. arXiv:math/9901072.

[Mar07] E. Markman, *Integral generators for the cohomology ring of moduli spaces of sheaves over Poisson surfaces*, Adv. Math. **208** (2007), no. 2, 622–646. doi:10.1016/j.aim.2006.03.006.

[MM25] D. Mattei and R. Meinsma, *Obstruction classes for moduli spaces of sheaves and Lagrangian fibrations*, preprint, 21 July 2025. arXiv:2404.16652v2.

[Mor84] D. R. Morrison, *On K3 surfaces with large Picard number*, Invent. Math. **75** (1984), 105–121. doi:10.1007/BF01403093.

[Son14] J. Song, *Riemannian geometry of Kähler–Einstein currents*, preprint, 2014. arXiv:1404.0445.

[Tos09] V. Tosatti, *Limits of Calabi–Yau metrics when the Kähler class degenerates*, J. Eur. Math. Soc. (JEMS) **11** (2009), no. 4, 755–776. doi:10.4171/JEMS/165.

[Wue26a] B. J. Wuebben, *Quantitative metric wall crossing for K3 bundle moduli*, 6 November 2026. doi:10.5281/zenodo.24024998.

[Wue26b] B. J. Wuebben, *Quantitative local models for Hermitian–Einstein metrics*, 7 November 2026. doi:10.5281/zenodo.24029090.

[Wue26c] B. J. Wuebben, *Exceptional cycles and harmonic metrics in K3 bundle moduli*, 7 November 2026. doi:10.5281/zenodo.24046092.

[Yos00] K. Yoshioka, *Irreducibility of moduli spaces of vector bundles on K3 surfaces*, preprint, 2000. arXiv:math/9907001v2.

[Yos01] K. Yoshioka, *Moduli spaces of stable sheaves on abelian surfaces*, Math. Ann. **321** (2001), 817–884. doi:10.1007/s002080100255.

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